

Understanding β : Physical vs. Linear Model

You are absolutely correct that physically, $\beta = 0.6$ for a standard open channel. Here is the mathematical proof of your insight, followed by why we must set $\beta = 1$ for a *linear* model.

1. The Physical Reality: Non-Linear Kinematic Wave ($\beta = 0.6$)

According to Manning's equation (KN Dulal Notes, pg. 16):

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} = \frac{1}{n} \frac{A^{5/3}}{P^{2/3}} S^{1/2}$$

Rearranging to solve for Area (A):

$$A = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{3/5} Q^{3/5}$$

Comparing this to the general kinematic momentum equation $A = \alpha Q^\beta$, we see exactly that:

$$\alpha = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{0.6} \quad \text{and} \quad \beta = \frac{3}{5} = 0.6$$

Because $\beta \neq 1$, the true kinematic wave is **Non-Linear**.

2. The Mathematical Simplification: Linear Kinematic Wave ($\beta = 1$)

The board exam question explicitly asked for the "**Linear Kinematic Wave Model**". A linear differential equation requires that variables are not raised to any power other than 1. Therefore, we make a deliberate mathematical assumption that Area is directly proportional to Discharge ($\beta = 1$).

$$A = \alpha Q^1$$

This simplification turns the wave celerity $c = \frac{dQ}{dA} = \frac{1}{\alpha}$ into a **constant**, allowing us to derive the simple FTFS explicit scheme without needing complex Newton-

Raphson iterations.

Q. Derive the space-time discretization expression of a linear kinematic wave model using forward difference in space and time.

Step 1: The Linear Governing Equation

Based on the kinematic wave approximation, the conservation of mass (continuity equation) without lateral inflow ($q = 0$) is:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

For a **linear** model, we assume $A = \alpha Q$ (where $\beta = 1$). Taking the partial derivative with respect to time:

$$\frac{\partial A}{\partial t} = \alpha \frac{\partial Q}{\partial t}$$

Substituting this back into the continuity equation gives:

$$\alpha \frac{\partial Q}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

Let the constant kinematic wave celerity be $c = \frac{1}{\alpha}$. Dividing the entire equation by α yields the standard linear kinematic wave equation:

$$\frac{\partial Q}{\partial t} + c \frac{\partial Q}{\partial x} = 0$$

Step 2: Forward Difference Approximations (FTFS)

We discretize the domain into a grid with space steps Δx (index i) and time steps Δt (index n). We approximate the continuous derivatives using Forward Differences:

$$\text{Forward Time (FT): } \frac{\partial Q}{\partial t} \approx \frac{Q_i^{n+1} - Q_i^n}{\Delta t}$$

$$\text{Forward Space (FS): } \frac{\partial Q}{\partial x} \approx \frac{Q_{i+1}^n - Q_i^n}{\Delta x}$$

Step 3: Substitution and Rearrangement

Substitute the approximations into the linear kinematic wave equation:

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta t} + c \left(\frac{Q_{i+1}^n - Q_i^n}{\Delta x} \right) = 0$$

Because this is an explicit scheme, we solve for the unknown future variable Q_i^{n+1} entirely in terms of the known present values at time step n . Multiplying by Δt :

$$Q_i^{n+1} - Q_i^n + c \frac{\Delta t}{\Delta x} (Q_{i+1}^n - Q_i^n) = 0$$

Step 4: Final Discretization Expression

Introducing the Courant number, $C_r = c \frac{\Delta t}{\Delta x}$, and isolating Q_i^{n+1} on the left side yields the final explicit FTFS discretized expression:

$$Q_i^{n+1} = Q_i^n - C_r (Q_{i+1}^n - Q_i^n)$$

or, factored by Q :

$$Q_i^{n+1} = (1 + C_r) Q_i^n - C_r Q_{i+1}^n$$